

COMP4431 Artificial Intelligence

Quiz 02 Solution

I. Knowledge-based Agent and Expert System

1.

a) Perform Forward Chaining as below:

Given Facts:

- The car won't start.
- The headlights are dim.

Step 1

Since the headlights are dim, **Rule 3** is triggered: i.e. "IF the headlights are dim THEN the battery is dead."

New fact derived:

- "The battery is dead."

Step 2

With the facts "The car won't start" and "The battery is dead," **Rule 1** is triggered: i.e. "IF the car won't start AND the battery is dead THEN the issue is a dead battery."

New fact derived:

The issue is a dead battery

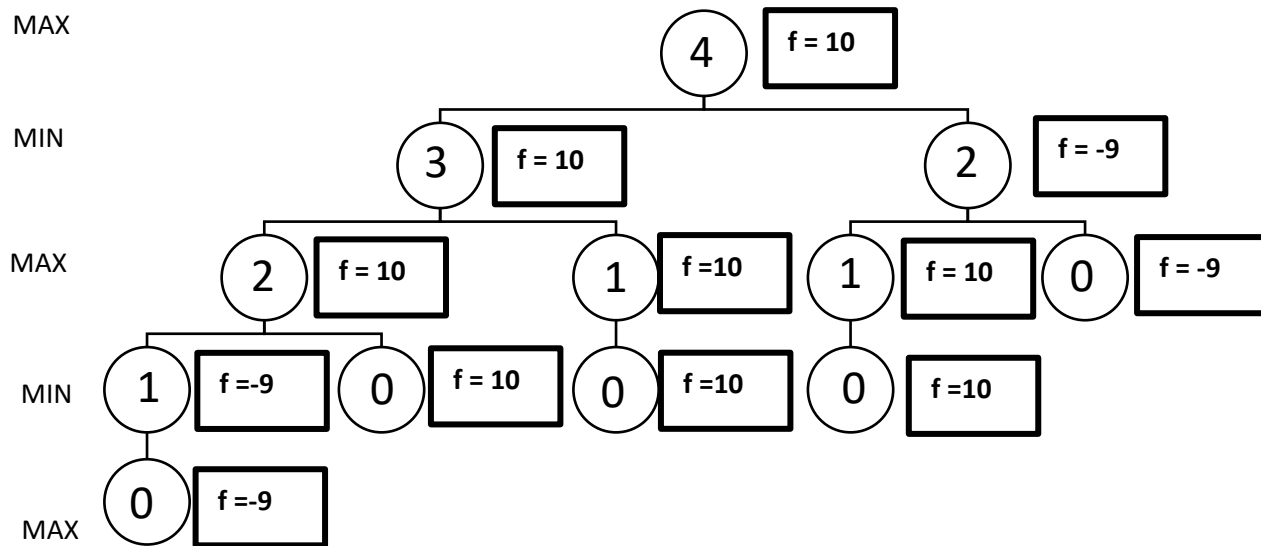
As no further rule can be triggered, we conclude that: The issue is a dead battery.

b) Another reasoning method used in expert systems is **backward chaining**.

Backward Chaining is a goal-driven approach which is the major difference to forward chaining. It **starts with a specific goal and works backward to determine if there is evidence to support it**. It works from the "top down," starting with the goal and searching for supporting facts. (or similar answer)

II. Game

2. a) The completed search tree of minimax algorithm is as follow:



The final decision is to take 1 stone on the first move.

c) Yes, the Minimax algorithm can still be used.

The major change is redefining the objective/utility function to reflect the new win/loss condition:

$f = -9$ if starting player takes the last stone and wins the game,

$f = 10$ if starting player does not take the last stone and loses the game

(or other acceptable answers)